

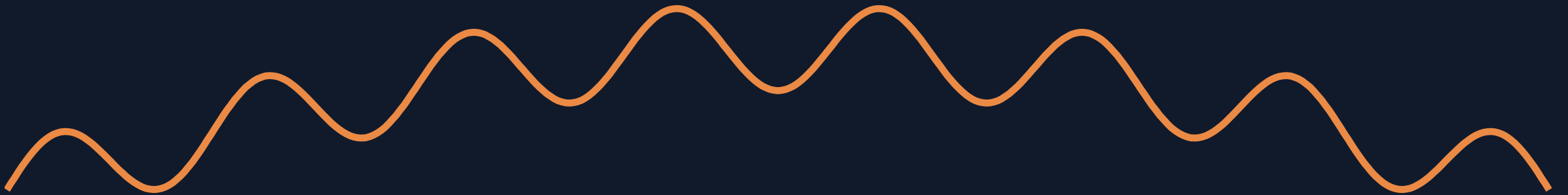
WISER SUMMER PROGRAM 2026

· BQP INDUSTRY CHALLENGE

Qapiino

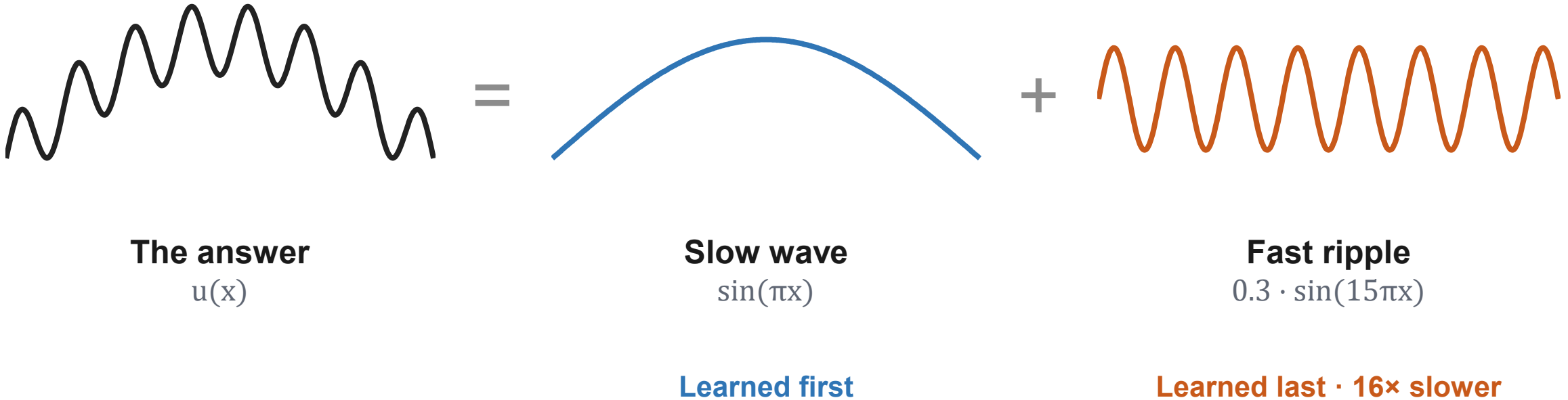
Letting the physics equation design the quantum circuit

Katta Lahari · Mannava Daasaradhi



QAPINN-Spectra · Spectrum-Matched Quantum-Assisted Physics-Informed Neural Networks

The problem: networks learn the fast ripple last



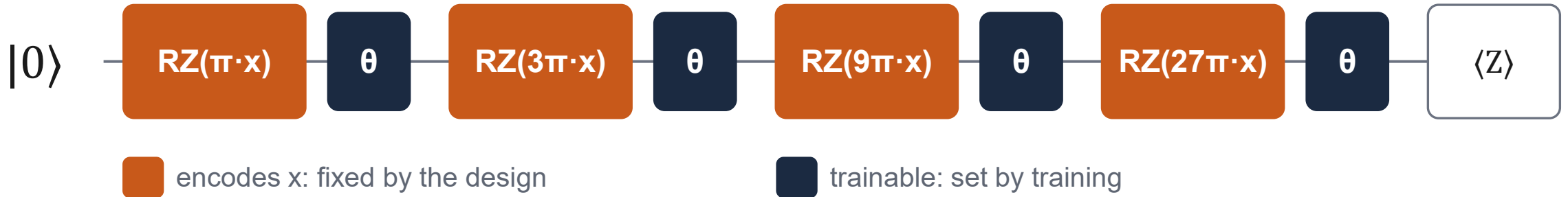
Why it matters

High-frequency problems are the ones engineers care about: sound, radar, shock waves.

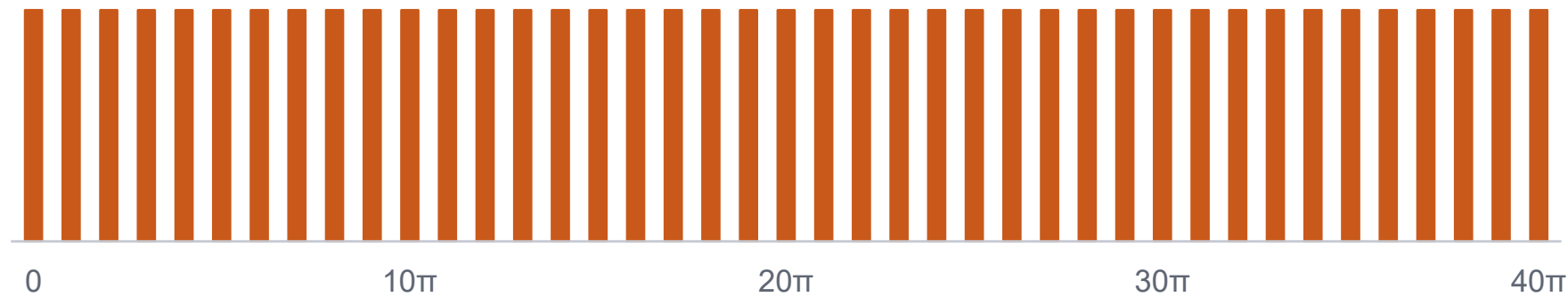
The open question

Which quantum circuit? Today it is picked by hand or by search, and every trial costs quantum hardware time.

The key idea: a quantum circuit is a set of waves



Waves it can make: every multiple of π up to $\pi + 3\pi + 9\pi + 27\pi = 40\pi$



41

waves
from 14 parameters

Why powers of 3? Like balance weights of 1, 3, 9 and 27 kg, which weigh every whole number up to 40: every wave, no gaps, no repeats. The encoding fixes which waves exist; training sets how loud each one is.

Our approach: the equation picks the circuit

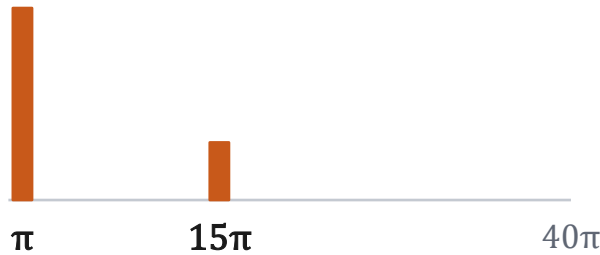
1 The equation

$$-u'' = f(x)$$

$f(x)$ contains $\sin(\pi x)$ and $\sin(15\pi x)$



2 Waves it needs



3 Depth by formula

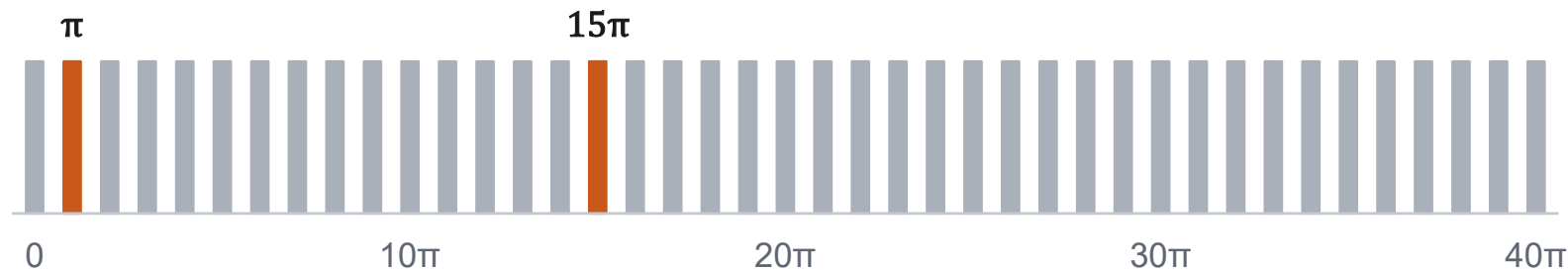
$$L = \lceil \log_3(2 \cdot 15 + 1) \rceil = 4$$

4 layers

scalings $\pi, 3\pi, 9\pi, 27\pi$

3 layers would stop at 13π and miss the ripple

The designed circuit's waves **cover both needed waves**



Covered.

No architecture search.

We give the circuit the resolution, not the answer. Only the top wave (15π) and the spacing (π) go in, like choosing a simulation's grid size from the physics.

What we built, and why

SMCD design rule

Equation in; circuit plus an auditable design card out. No search.

Own quantum simulator

Exact, in PyTorch; matches PennyLane to within 10^{-10} .

7 models × 6 problems

4 quantum, 3 classical, including a classical model handed our exact waves.

6 explainability tools

Same checkpoints on every model: how fast each wave is learned.

12 predictions, locked first

Committed to git before the first run; a script issues every verdict.

384 automated tests

Every figure and number regenerates from committed results.

Why these six problems

Poisson

clean slow wave + ripple

Heat

the no-go control

Burgers

nonlinear shock front

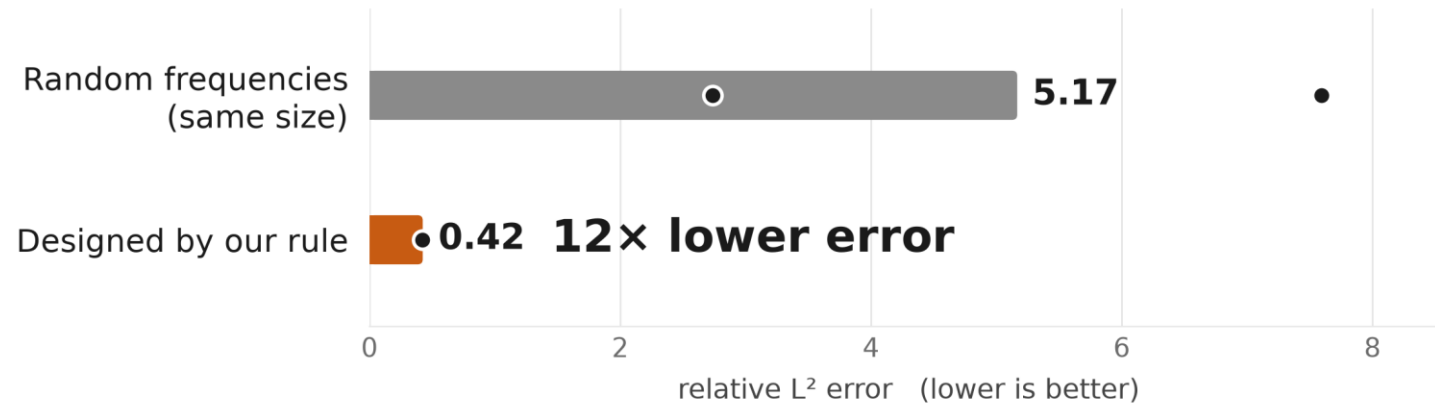
Helmholtz $k = 4, 10, 20$

a frequency ladder

Result: a designed circuit vs. a random one

Same size, same equation: the design is what makes it work

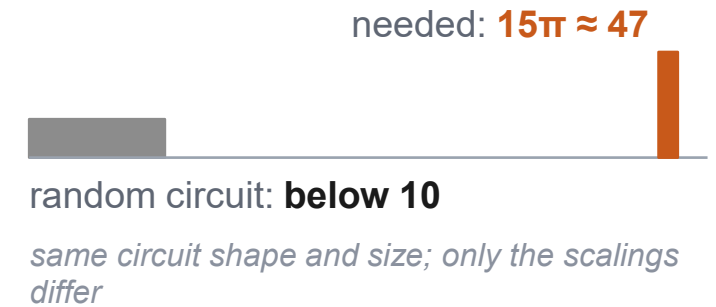
Heat equation · both quantum circuits have 18 trainable parameters



Bars: median of the 2 analysed seeds · dots: each seed · source: committed results/runs/ (PR-6 run selection)

Heat equation. Same size (18 parameters), same training; only the scalings differ. Both training runs agree.

Why the design wins

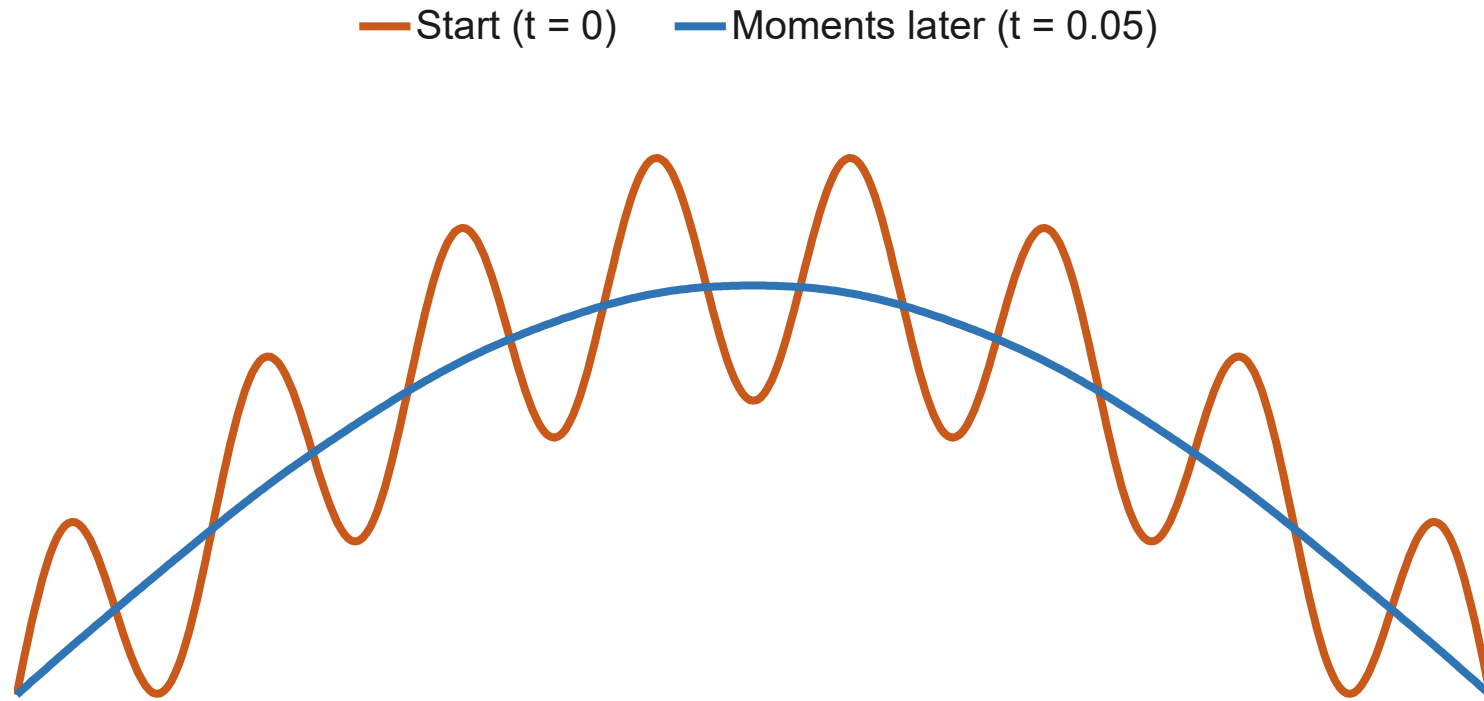


What you design is what you get



Measured output has exactly the designed waves; training barely moves them (limit set in advance).

Knowing when NOT to use quantum



The fast ripple fades like

$$e^{-111t}$$

gone by $t \approx 0.05$

No-go, predicted before any run

Predicted benefit: 2.5%. Three of our four quantum variants did worse than a standard network, as predicted.

Heat equation · exact solution

Limits, what we learned, and what's next

What we learned

OUR QUANTUM FINDING

Designing quantum circuits beats guessing them.

12× lower error than a same-size random circuit, with no search.

A TRAP WE CAUGHT

The textbook starting state switches off a third of the waves.

We found it by measuring the circuit's output, and fixed it.

THE INSIGHT

The waves alone aren't the quantum edge.

A classical model handed the same waves, at the same size, still does better.

Limits

- Simulated, not yet on quantum hardware
- 2 training runs per model in the final analysis
- Small test equations (1-D and 2-D)
- Assumes the needed waves can be read from the equation

Next

- 5 training runs per model
- Equations with too many waves to hand to a classical model
- Real quantum hardware

Why it matters

Design rule

Equation in, circuit out. No trial and error.

Go / no-go

Know where quantum won't pay off, before spending hardware time.

Pre-registered test

A reusable, auditable check for any quantum-PINN claim.

Where it could apply: wave problems in acoustics, radar and electromagnetics, and flows with sharp fronts.

Takeaway: the equation can pick its own quantum circuit, with no search, and that pick is what makes the circuit work.

Thank you

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